Appendix — Mixture-of-Experts Upgrade
Short / Long / VeryLong + Chat / Casual / Essay

This appendix is merged into the main report. It adds small-text datasets, length-specialized experts, a text-type gate, and threshold calibration.

Code: `moe/download_small_text.py, moe/train_moe.py, moe/predict_moe.py, moe/calibrate_thresholds.py`

Artifacts: `models/moe/expert_{short,long,verylong}.pkl, type_gate.pkl, thresholds.json`

A Motivation — Short-Text Failure

The Phase-1 baseline scores 99%+ on full essays but collapses on short text: first 25 words → 67.4% (LR), first 50 → 83.3%, 1 sentence → 59.8%. Errors are one-sided: AI recall stays ~0.98 while human false-positives explode (~51% of humans called AI at 25w). This is expected — training never saw short text (train.csv median 363w, 5th percentile 180w) and TF-IDF + stylometrics need length.

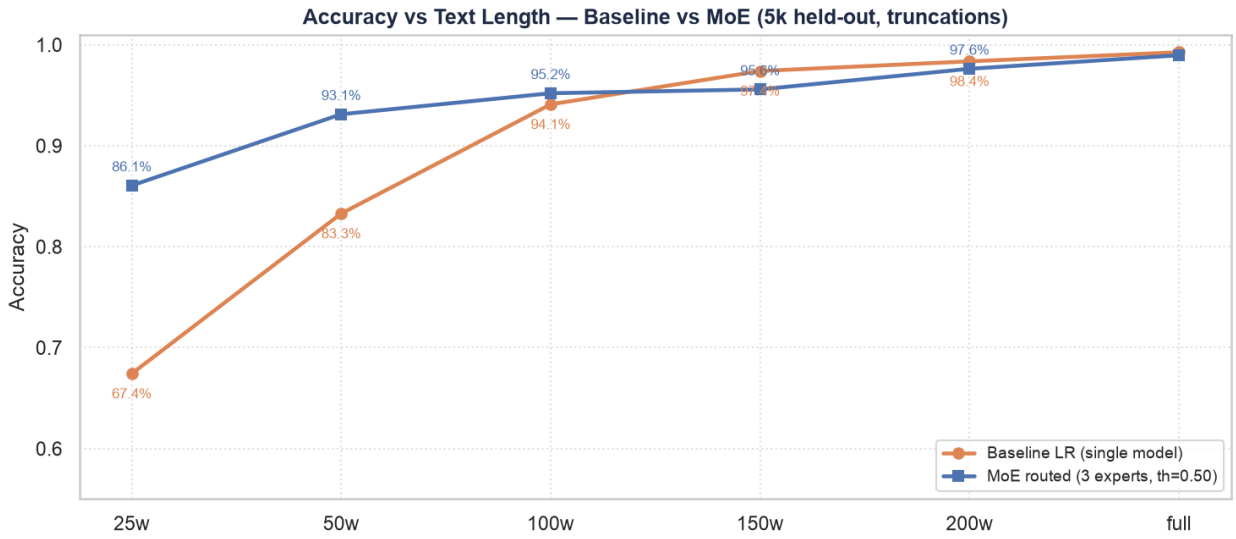


Figure A1: Accuracy vs truncated length (5k held-out). MoE lifts 25w 67%→86% and 1-sentence 60%→82%.

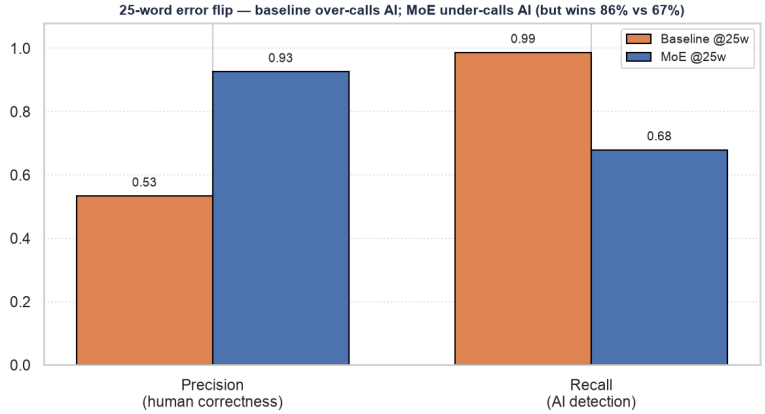


Figure A2: Error flip at 25w. Baseline over-calls AI (P=0.53, R=0.99); MoE under-calls AI (P=0.93, R=0.68) — net acc still wins.

The baseline also fails cross-domain: 48.6% on unseen HC3 chat answers (worse than chance) and 85.5% on modern GPT-4/Claude essays. It learned old-essay quirks, not authorship.

B New Datasets (Small Text + Modern Essays)

Downloaded via `moe/download_small_text.py` (JSONL/parquet directly from Hugging Face Hub):

Source	n	Median	Role
HC3 reddit_eli5+open_qa+finance (Hello-SimpleAI/HC3, Guo et al.)	80,811 (56k human / 25k AI)	117w, 61% ≤150w	SHORT chat/casual
artnitolog/llm-generated-texts (GPT-4 Turbo/Omni, Claude 3 Opus, Llama3 70B, Command R+)	17,997 (3k human / 15k AI)	510w, 68% >400w	LONG/VERYLONG modern
Existing train.csv 20k balanced + truncation augmentation	20,000 essays + 20k trunc copies	393w avg; trunc 50/100/150/200/300 w	LONG/VERYLONG + short-essay style

Table A1: New data. HC3 gives genuine short Q&A; (human vs ChatGPT); modern essays give 2024-LLM text.

Truncation augmentation: each train.csv essay contributes a prefix or random 50/100/150w copy (label preserved) so the short expert learns short-essay style, not just chat. Pool caps are balanced per class (seed 42): short 20k, long

~16.9k, verylong ~11.4k.

C MoE Architecture

Length gate (deterministic): word count ≤150 → **short**, 151-400 → **long**, >400 → **verylong** (150 = observed accuracy cliff, 400 ≈ dataset median). **Each expert** has its own Word TF-IDF (1,2) 15k + Char TF-IDF (char_wb 3,5) 25k + LogisticRegression(C=1.5, lbfgs) — same hyper-parameters as the baseline, but features specialized per length regime.

Type gate (learned): LogisticRegression over 16 fast stylometric features (log words, sentence stats, TTR, hapax, contraction/informal/first/second-person rates, punctuation rates, FRE/FKGL, burstiness) → **chat / casual / essay**. Labels from source: HC3 `open_qa` → chat, HC3 eli5/finance + truncated essays → casual, full/modern essays → essay. Trained on 7,500 balanced rows (2,500/type); validation acc 80.5%.

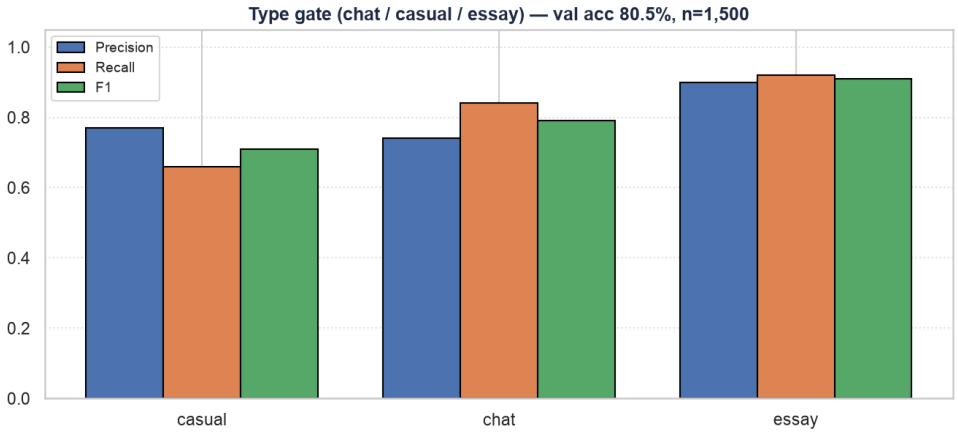


Figure A3: Type gate per-class scores. Essay easiest (0.91 F1); casual/chat confuse (casual recall 0.66), as expected for informal text.

Inference (`moe/predict_moe.py`) routes by length, reports label + P(AI) + expert + word count + type (+confidence) + reliability warning (<25w / ≤50w).

D Per-Expert Results

Expert	Train / Val	Acc	F1	AUC
short	16,000 / 4,000	0.9700	0.9700	0.9960
long	13,491 / 3,373	0.9819	0.9832	0.9982
verylong	9,137 / 2,285	0.9842	0.9821	0.9955

Table A2: Per-bucket validation (results/moe_expert_metrics.csv).

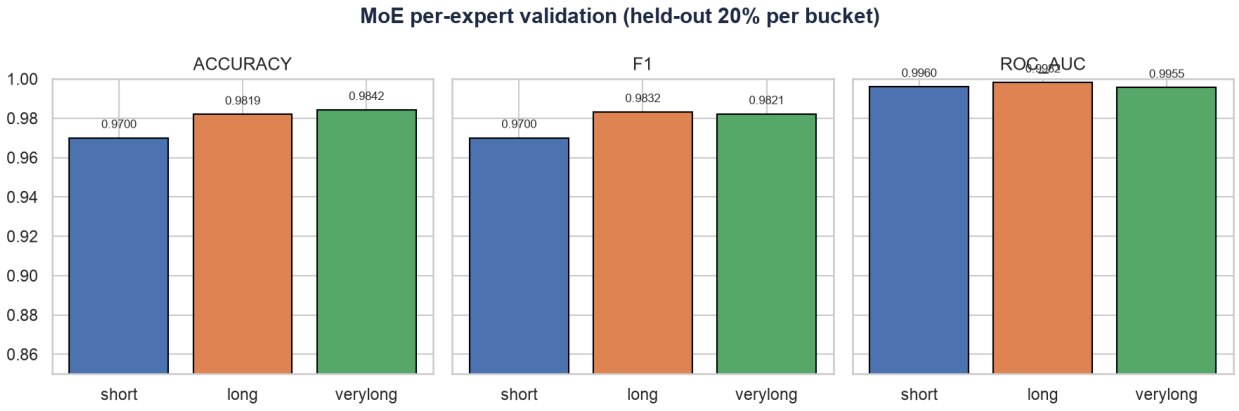


Figure A4: Per-expert accuracy/F1/AUC (all ≥0.97).

E Routed MoE vs Baseline (5k held-out + new domains)

Routing = length → expert. Full-text MoE costs 0.3pp vs baseline (98.96% vs 99.26%) — the price of specialization. Everywhere else it wins, and on unseen domains the gap is decisive (HC3: 48.6%→98.5%, modern: 85.5%→95.7%).

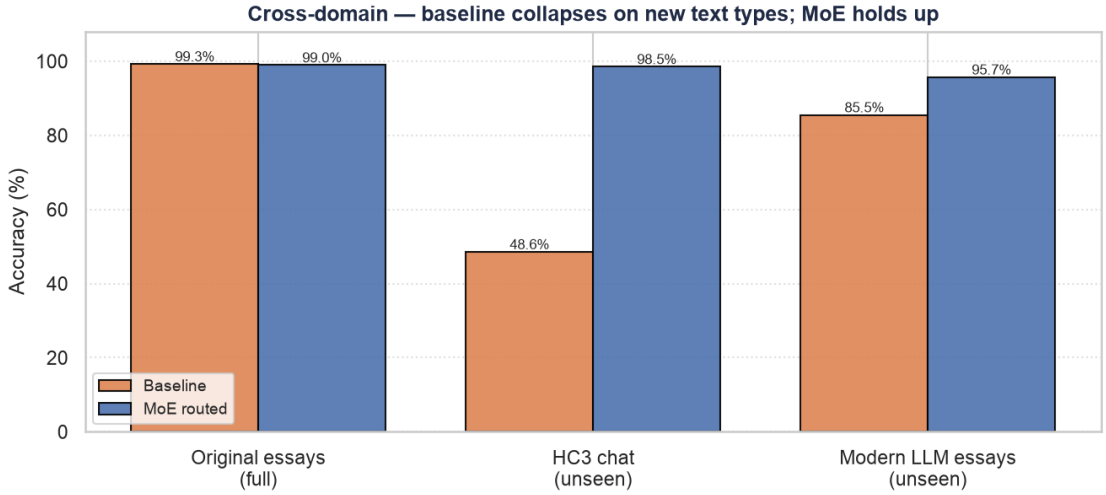


Figure A5: Cross-domain held-outs. Baseline is worse than chance on HC3 chat.

Condition	Baseline acc	MoE acc	Delta
full	99.26%	98.96%	+-0.3pp
first 25w	67.42%	86.06%	+18.6pp
first 50w	83.28%	93.12%	+9.8pp
first 1 sent	59.82%	81.64%	+21.8pp
HC3 holdout	48.56%	98.52%	+50.0pp
modern holdout	85.52%	95.70%	+10.2pp

Table A3: Headline comparison (results/moe_routed_metrics.csv + moe_baseline_comparison.csv).

Per-bucket on full text: verylong 99.59% (n=1,958), long 98.81% (n=2,945), short 90.72% (n=97 naturally-short essays).

F Threshold Calibration

The short expert under-calls AI on ultra-short essays (recall 0.68 at 25w). Calibrating on a calib half (2,500, seed 999, max F1) gave per-length thresholds 25w→0.32, 50w→0.40, 100w→0.56, 150w→0.64, implemented as `th=0.28+0.0024·n` clamped [0.32,0.64] for the short bucket (`models/moe/thresholds.json`). Long/verylong stayed at 0.50.

Eval-2500 condition	th=0.50	Adaptive	Note
first 25w	86.16% / F1 0.785	88.08% / F1 0.837	Recall 0.68→0.82
first 50w	93.12% / 0.905	93.08% / 0.909	Neutral (R↑ P↓)
first 100w	94.52% / 0.928	94.52% / 0.928	th≈0.52, identical
first 150w	94.96% / 0.935	96.20% / 0.949	+1.2pp
full	98.96%	98.92%	Flat

Table A4: Calibration on unseen eval half (results/moe_calibrated_metrics.csv). Caveat: HC3 natural short –0.6pp (human-heavy); essays gain. The <50w warning is retained.

Demo flips: a 31w borderline excerpt (P(AI)=0.374, th=0.354) is now AI-Generated (was Human under 0.50) — the intended fix, still low-confidence and flagged. A 14w casual human (P(AI)=0.035) stays clearly human.

G Discussion & Reproducibility

- **What MoE fixes:** length specialization + genuine short data. HC3 recall was the unlock — the baseline had never seen chat Q&A.; The 20k→16k per-expert vocabulary (vs 75k joint) also avoids long-essay features drowning short text.

- **What remains hard:** <25-50w is intrinsically weak (even MoE 86-93% with warnings). 150w essays still prefer the baseline 97.4% vs 95.6% — the short expert leans chat; blending experts at the boundary is future work.
- **Thresholds transfer:** calibration helps essay truncations but slightly hurts human-heavy HC3 natural short — thresholds are domain-sensitive; keep them length-adaptive, not global.
- **Type gate:** 80.5% is enough for routing insight, not for enforcement. Casual↔chat confusion is linguistic, not a bug.

Reproduce: `PYTHONPATH=. python3 moe/download_small_text.py` → `moe/train_moe.py` → `moe/calibrate_thresholds.py` → `moe/make_moe_appendix.py`. All artifacts under `models/moe/`, metrics under `results/moe_*.csv`.

Appendix References

- Guo, B. et al. How Close is ChatGPT to Human Experts? HC3 / Hello-SimpleAI/HC3. (2023).
- artnitolog/llm-generated-texts (GPT-4 Turbo/Omni, Claude 3 Opus, Llama3, Command R+ essay prompts).
- Roy, R. et al. A Comprehensive Dataset for Human vs AI Generated Text Detection (Defactify). arXiv:2510.22874 (2025) — context for cross-domain eval.
- Baseline report references apply (Hastie et al.; Robertson & Zaragoza; Sentence-BERT/DistilBERT/RoBERTa).